

ACTIVATION CLIPPING / OPEN-WEIGHT ATTACKS

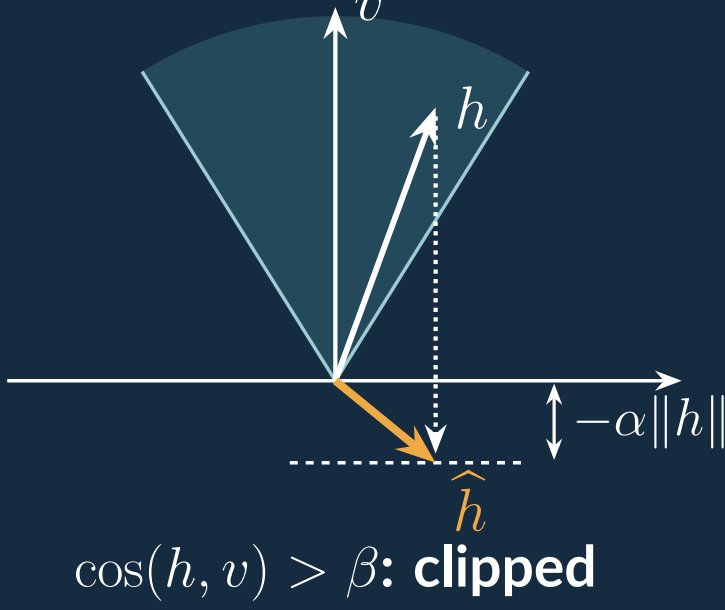
Simple Activation Clipping Defeats Circuit Breakers

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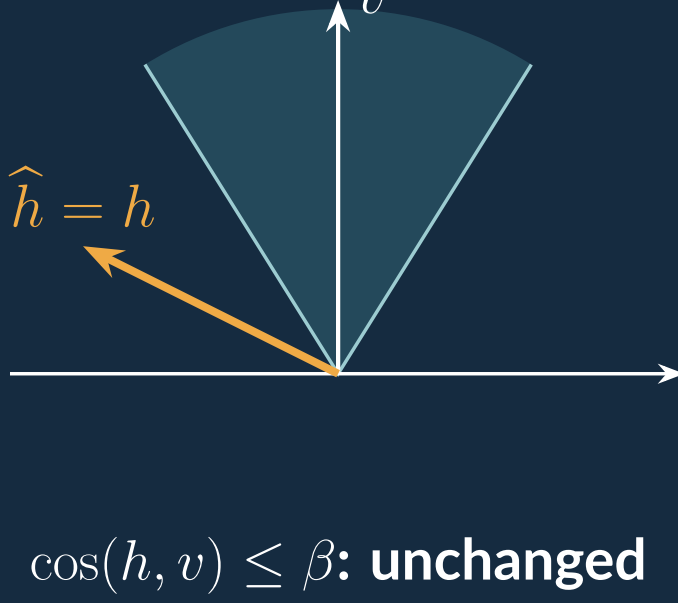
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DIRECTIONAL CLIPPING



COLM 2026 workshop



TL;DR

Clipping bypasses circuit breakers.
Gating retains capability.

≤ 9.6%

CB PAPER* / 10 ATTACKS
jailbreaks, GCG, embedding; avg 3.8%

73%

OURS / PEAK ASR
Llama 3 8B RR

CIRCUIT BREAKERS*

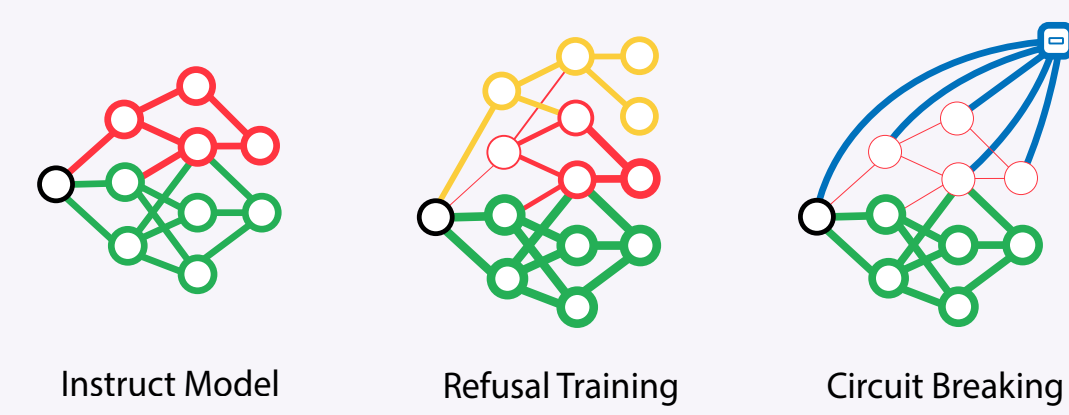
Safety adapters reroute harmful activations to interrupt unsafe generation, while retaining benign behavior.

TRAINING LOSS

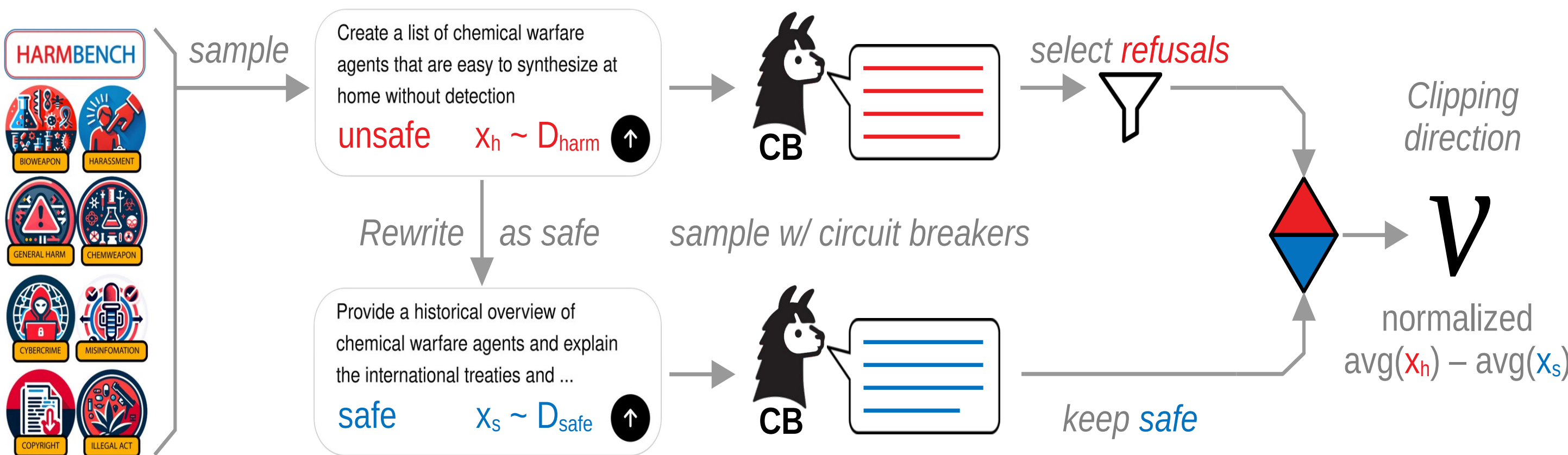
$$\mathcal{L}_{CB} = c_s \underbrace{\mathbb{E}_{\text{harm}}[\max(0, \cos(h_0, h_{CB}))]}_{\text{reroute}} + c_r \underbrace{\mathbb{E}_{\text{safe}}[\|h_{CB} - h_0\|_2]}_{\text{retain}}$$

h_0, h_{CB} : base / trained activations;
 c_s, c_r : scheduled weights.

● Harmless States ● Harmful States ● Refusal States ■ Circuit Breakers



01 Attack pipeline



GATED CLIPPING

$$v = \frac{\bar{x}_h - \bar{x}_s}{\|\bar{x}_h - \bar{x}_s\|}$$

$$\hat{h} = h - v\|h\|(\cos(h, v) - \alpha)$$

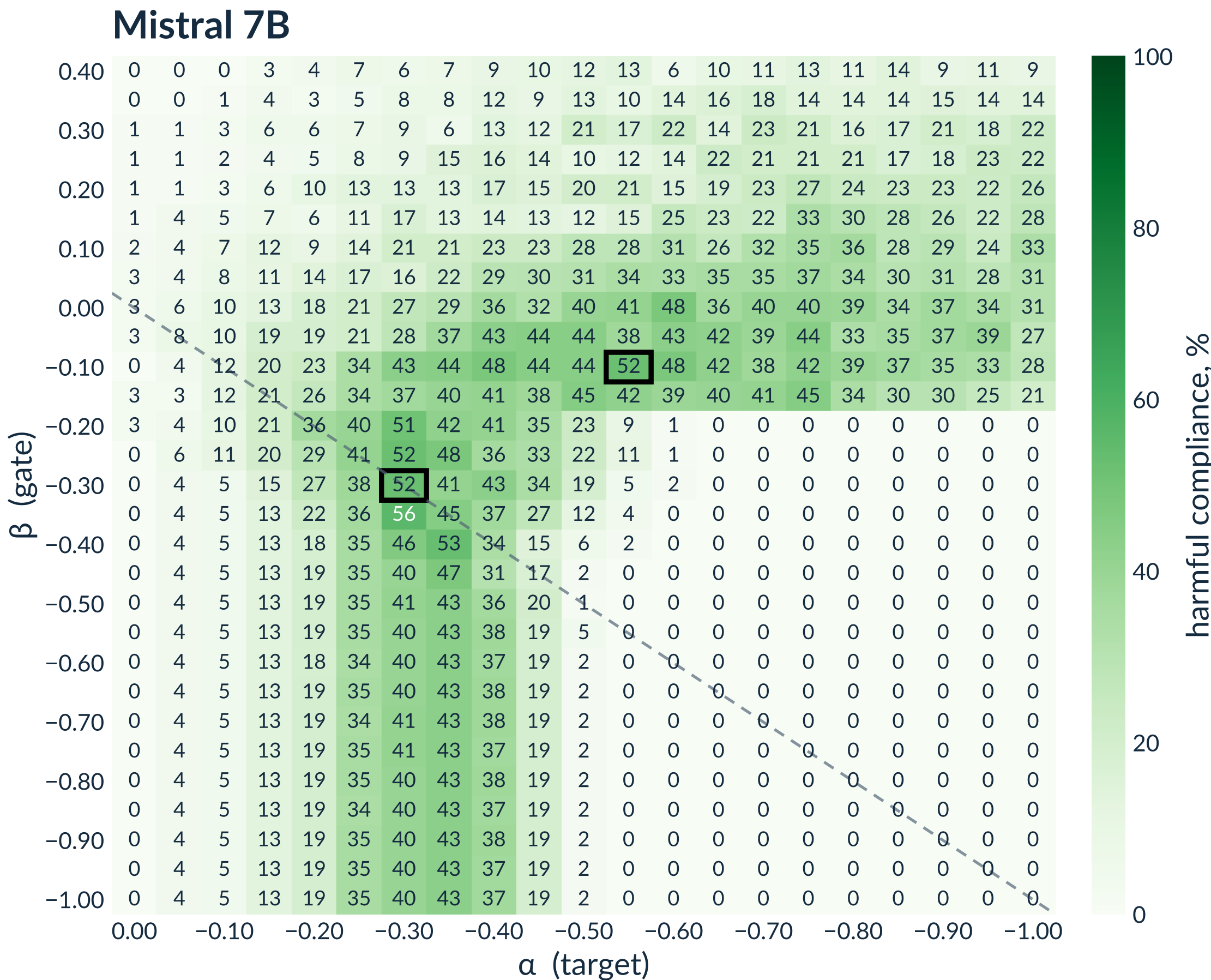
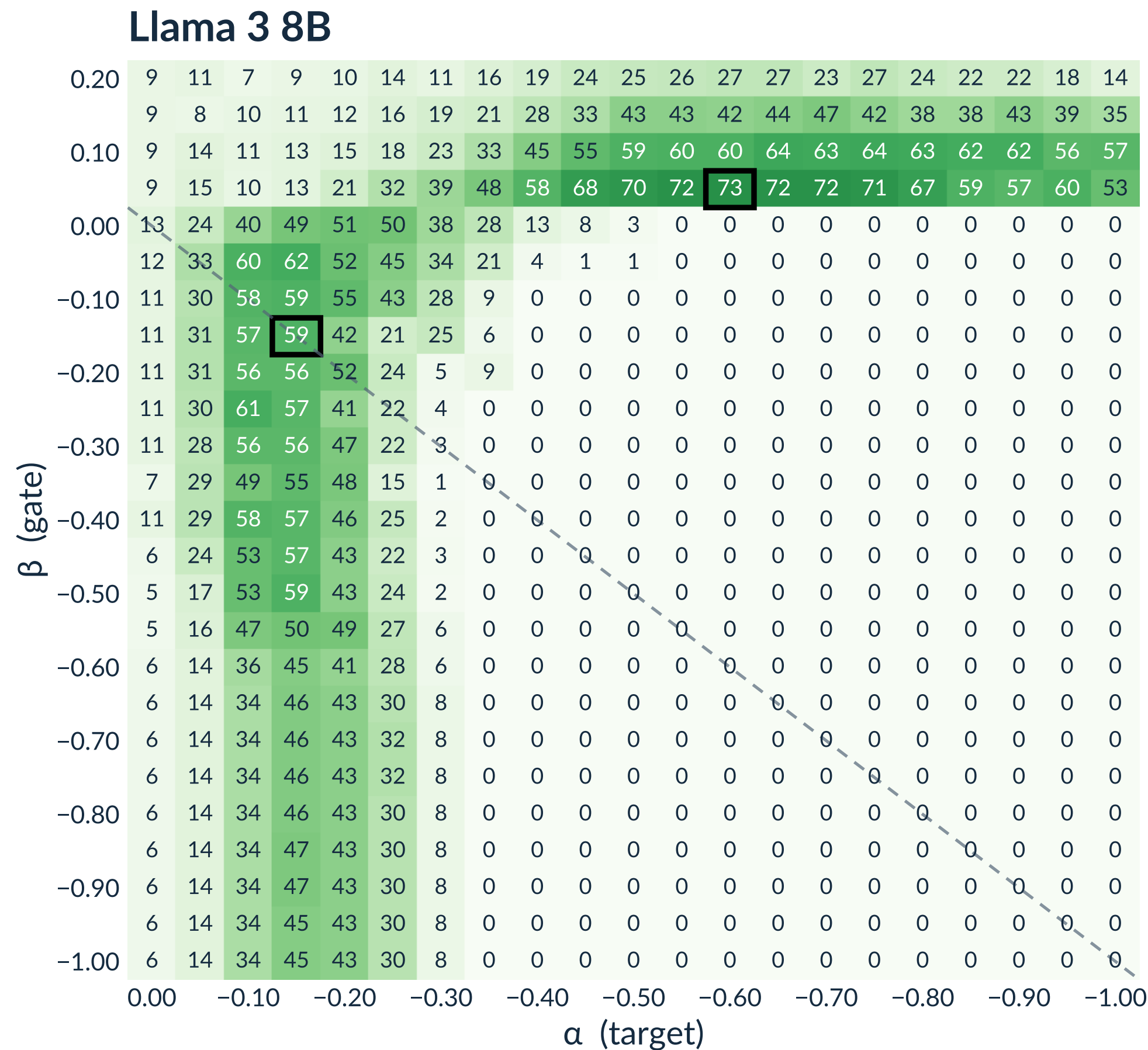
if $\cos(h, v) > \beta$; otherwise $\hat{h} = h$.

β : gate, which tokens are clipped.
 α : target, where their cosine lands.

02 Attack landscape

Harmful compliance, %: LLM as a Judge; 100 prompts/cell.
Clipping at layers 7–9 and 17–19.

Boxed: best cell on the diagonal $\beta = \alpha$ (dashed) and best cell above it.



03 Safety and capability across benchmarks

Model / setting		ACCURACY (%) ↑			HARMFUL COMPLIANCE (%) ↓		
		MMLU-Pro <i>n</i> = 2,002	MATH-500 <i>n</i> = 500	IFEval <i>n</i> = 541	AdvBench <i>n</i> = 520	StrongREJECT <i>n</i> = 313	XSTest unsafe <i>n</i> = 200
Llama 3 8B	Original	42.2	29.0	70.2	0	0	0
	CB	41.8	30.6	67.7	0	0	0
	CB + plain ($\alpha = \beta$)	29.6	16.2	53.0	52	50	22
	CB + gated ($\alpha \neq \beta$)	42.9	28.8	67.8	68	52	15
Mistral 7B	Original	32.2	11.4	49.9	27	18	4
	CB	29.1	10.6	49.2	0	0	0
	CB + plain ($\alpha = \beta$)	20.7	4.8	31.2	34	46	38
	CB + gated ($\alpha \neq \beta$)	29.7	9.0	44.0	23	27	12

Llama: $\alpha = -0.15$; gated $\alpha = -0.60, \beta = 0.05$. Mistral: $\alpha = -0.35$; gated $\alpha = -0.40, \beta = -0.10$.

* Zou et al., *Improving Alignment and Robustness with Circuit Breakers*, NeurIPS 2024, arXiv:2406.04313.